Mukul DHIMAN

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Research Interests / MOTIVATION

- Strong motivation to develop multiscale theoretical frameworks linking ultrafast excitonic processes to environmental dynamics, with a focus on energy dissipation and quantum friction at solid-liquid interfaces in low-dimensional materials.

- First-principles modelling of excited states and excitons in condensed-phase and interfacial environments, electronic structure methods (CASPT2, MRCI, DFT), non-adiabatic molecular dynamics and study of open-quantum systems using hybrid MD methods to investigate exciton–solvent coupling, nanoscale dissipation and interfacial phenomena.

- Expertise in excited-state and non-adiabatic dynamics, including surface hopping algorithms, mixed quantum-classical methods, and wave packet propagation for energy transfer at surfaces and interfaces.

- Demonstrated experience with MD incorporating nuclear quantum effects (adQTB, PIMD) in solution-phase systems, and DFT-based calculation of surface potentials and phonon modes using Quantum ESPRESSO.

- Proven capability in developing in-house scientific codes (C, Fortran, Python) and working on HPC clusters, with a track record of rapid integration across five research institutions in four countries.

- Interested in physics-informed and data-efficient AI: embedding physical constraints into learning models to improve reliability, interpretability, and scalability.

- Comfortable working across disciplines (physics, chemistry, materials science) and rapidly learning new formalisms, codes, and computational frameworks.

- Motivated to apply scientific AI to advanced materials and energy-related systems, including characterization, modeling, and performance prediction.

RESEARCH EXPERIENCE

Independent Researcher March'26- Present

- Currently working on building a new hybrid model with quantum thermal bath while accounting nuclear quantum effects on diffusion dynamics

MSME-CT, Universite Gustav Eiffel (1 Published Article)

Postdoc researcher Sept'24-March'26

- *Ab initio* calculation to obtain electronic structure of Acrolein molecule (MOLCAS and GAUSSIAN)
- Using **adaptive Quantum Thermal Bath dynamics (adQTB) MD** and **Path Integral MD (PIMD)** to study absorption spectrum of gas phase Acrolein and in water to include nuclear quantum effects (NQEs)
- Force Field Parametrisation and validation
- Developed plethora of python codes to extract and calculate absorption spectrum.
- Teaching L1 and Supervising M1 students

ISMO(SYSTEMAE), CNRS, University Paris Saclay (1 Published Article)

Postdoc researcher May'23-Jun'24

- Utilisation of Mixed Quantum Classical model to study hydrogen sticking on graphene surface with temperature, **calculating sticking probabilities** for both physisorption and chemisorption potentials
- Used Quantum ESPRESSO to obtain potential and phonons modes
- **Mixed Quantum Classical Dynamics** and sticking probabilities are done using in-home code based on **closely coupled wave packet dynamics (CCWP)**
- Quantum Dynamics and sticking using **Hybrid Trajectory (HyTRAJ)** method based on Smolik Grid method

Goethe University Frankfurt am Main

Postdoc researcher Oct'22-April'23

- Modelling Singlet Fission process using MCTDH quantum chemistry code
- Teaching Theoretical Chemistry practical sessions

CIMAP, GANIL-CEA, CNRS, University of Caen (2 Published Articles)

Doctorate in Physics Nov'18-June'22

- Algorithm **development and modelling of Non-Adiabatic Molecular Dynamics with Surface Hopping using C** to study collision, fragmentation dynamics and **diabatization effects**
- Semi Empirical, HPP and DIM models were developed using C
- *Ab initio* (RASSCF, CASPT2, MRCI) **Electronic Structure calculations using MOLCAS**
- Use of Mathematica for simple model simulations and validations

Peter the Great St. Petersburg Polytechnic University

Graduate Researcher Feb'17-July'18

- Study of Optical Properties of semiconductor Quantum Dots using RPA with local exchange developed on Fortran
- Performed calculation to determine the Energy dispersion of Quantum Dots
- Finite Element analysis of electronic waveguide in tetrahedral quantum dots using COMSOL

EDUCATION

Normandie University, UNICAEN **France**
PhD. Physics, Cluster and Solid Matter Physics Nov'18-May'22

Peter the Great St. Petersburg Polytechnic University **Russia**
M.S. Physics, Mesoscopic and Advanced Material Science Sept'16-July'18

Technical University of Munich **Germany**
M.S. Physics, Erasmus+ Exchange Semester Oct'17-March'18

Shivalik College of Engineering, UTU **India**
B.S. Electrical and Electronics Engineering Aug'09-July'13

ADDITIONAL EXPERIENCE

Magneti Marelli UM Electronics Delhi, India
Software Testing April'14-Aug'16

Coaching Institute Delhi, India
Physics Teacher for High Schoolers Aug'13-Mar'14

SCHOLARSHIPS/GRANTS

PhD Research Grant: NORMANDY UNIVERSITY

ERASMUS+ Grant for exchange

SKILLS

Computer Languages: C, C++, Python, Fortran

Hands on with Simulations Software: Tinker-HP, AMBER, Quantum Espresso, Mathematica, MOLCAS Quantum Chemistry Code, MCTDH Quantum Chemistry Software, Gaussian, Origin Pro, Gnuplot, LabView, MATLAB, Simulink.

Others: Developed Machine Learning algorithms, Multi-Thread Bash Scripting on Linux High Performance Clusters (HPC)

Language: French (B2 Level), Fluent in English (C1-C2 Level), Hindi (Native), Russian (A2) and German (A1)

HOBBIES

Sports: Endurance Cycling, Tennis, Badminton and Hiking
Extra Activities: Cooking, Painting, visiting museums and discussing art and culture.

PUBLICATIONS AND TALKS

Peer Reviewed Journal Article:

- Dhiman M, Navizet I, Huppert S & Mangaud E Simulating steady state spectra of Acrolein while accounting Nuclear Quantum Effects
  J. Phys. Chem. A 2026, 130, 26, 4900–4912

- Dhiman, M. & Gervais, B. Correcting Rydberg Excitations in Argon Clusters: From Trimer to Excimer Localization
  Chemical Physics Letters (2026) 142825

- Dhiman M. and Morriset S. Chemisorption and physisorption of an H atom on a graphenic surface: Interaction potential and sticking probabilities
  Chemical Physics Letters 871 (2025) 142104

- Dhiman, M., Douady, J., & Gervais, B. (2022). Geometry, absorption and luminescence of small Ar clusters.
  Molecular Physics, 120(9) 2022

Non-Peer Reviewed Articles:

- Dhiman. M (**In Preparation, 09/2026**) A Theoretical Model for Hybrid Quantum-Thermal Bath
(A Proof-of-Concept Study to model diffusion dynamics using Quantum Thermal Bath)
- Dhiman. M, Giglio. E & Gervais. B (**In preparation**) Excited State Dynamics on Neutral Argon Cluster: Exploration of Dissociation Channels

PhD Thesis: Dhiman, M. (2022). *Modelling excited argon clusters: geometries, spectroscopic properties, and non-adiabatic relaxation dynamics* (Doctoral dissertation, Normandie Université).
https://theses.hal.science/tel-03775595

MS Thesis: Dhiman, M. (2018). Surface plasmons in optical spectrum of ZnO nanocrystals: 10.18720/SPBPU/2/v18-1793

Seminars Presentations:

- **INSP 2025:** "Study of Energy and Charge Transfer Processes in Varied Systems"
- **LPCMR 2025:** "Modelling Excitation Relaxation with Quantum Transitions in neutral clusters: A pathway to Electron Dynamics and Quantum Coherences?"
- **ISMO 2024:** "Study of Energy and Charge Transfer Processes in Varied Systems"
- **Group Annika Bande 2024:** Study of Energy and Charge Transfer Processes in Clusters and Surfaces

Conference Presentations:

- Dhiman M *et al* Simulating steady state spectra of Acrolein while accounting Nuclear Quantum Effects GDR EMIE Toulouse 2026
- Dhiman M. Hydrogen Sticking Dynamics on Graphene Surface, Journées de chimie théorique du pôle IdF/Nord de ThéMoSiA 2025
- Dhiman, M & Gervais, B. (2024). *Excitation Dynamics with Quantum Transitions in Neutral Argon Clusters*, Hot Topic Presentation at SASP2024 Conference
- Dhiman, M., Douady, J., & Gervais, B. (2021). *On the structure and electronic spectra of excited neutral argon rare gas clusters using Hole Particle Pseudopotential method*, Grands instruments pour la physico-chimie moléculaire Conference Presentation
- Dhiman, M., Douady, J., & Gervais, B. (2021). *On the Electronic Structure of excited neutral Argon Clusters*, Pre-RGD32 online workshop, Korea

Poster Presentation:

- Dhiman, M, S. Morriset, N. Rougeau & D. Lauvergnat (2025). *Hydrogen Sticking Dynamics on Graphene Surface using Hybrid Trajectories*, ECAMP 15 Conference, Innsbruck
- Dhiman M, Mangaud E, Huppert S & Navizet I (2025) *Simulating steady state spectra of Acrolein while accounting Nuclear Quantum Effects*, ECAMP 15 Conference, Innsbruck
- Dhiman M, Mangaud E, Huppert S & Navizet I (2025) *Simulating steady state spectra of Acrolein while accounting Nuclear Quantum Effects*, Universite Gutave Eiffel Day (Best Poster Prize)
- Dhiman, M, S. Morriset, N. Rougeau & D. Lauvergnat (2024). *Hydrogen Sticking Dynamics on Graphene Surface using Hybrid Trajectories*, Poster Presentation at AMOC2024 Conference
- Dhiman, M, S. Morriset, N. Rougeau & D. Lauvergnat (2024). *Hydrogen Sticking Dynamics on Graphene Surface using Hybrid Trajectories*, Poster Presentation at SASP2024 Conference
- Dhiman, M, S. Morriset & N. Rougeau (2023). *Hydrogen Sticking Dynamics on Graphene Surface using Hybrid Trajectories*, Poster Presentation at Quantum Espresso Workshop Pavia
- Dhiman, M., Douady, J., & Gervais, B. Poster Presentation (2019) at Ecole EMIE-UP on Multiscale Dynamics in Molecular Systems in Ecole de Physique, Les Houches, Aug'19